

Problem Set 19

Problem 15.2.2:

(1): $\frac{1}{2} \otimes 1 = \frac{3}{2} \oplus \frac{1}{2}$, so $j_1 = \frac{1}{2}$ and $j_2 = 1$ implies $j \in \{\frac{3}{2}, \frac{1}{2}\}$. Since there is a unique state with $m = \frac{3}{2}$ we must have

$$(|jm\rangle \equiv) \left| \frac{3}{2} \frac{3}{2} \right\rangle = \left| \frac{1}{2} \frac{1}{2}, 11 \right\rangle \quad (\equiv |j_1 m_1, j_2 m_2\rangle). \quad (1)$$

Then, $J_- \left| \frac{3}{2} \frac{3}{2} \right\rangle = \hbar \sqrt{(\frac{3}{2} + \frac{3}{2})(\frac{3}{2} - \frac{3}{2} + 1)} \left| \frac{3}{2} \frac{1}{2} \right\rangle = \hbar \sqrt{3} \left| \frac{3}{2} \frac{1}{2} \right\rangle$, so

$$\begin{aligned} \left| \frac{3}{2} \frac{1}{2} \right\rangle &= \frac{1}{\hbar \sqrt{3}} J_- \left| \frac{3}{2} \frac{3}{2} \right\rangle = \frac{1}{\hbar \sqrt{3}} (J_{1-} + J_{2-}) \left| \frac{1}{2} \frac{1}{2}, 11 \right\rangle \\ &= \frac{1}{\sqrt{3}} \left(\sqrt{(\frac{1}{2} + \frac{1}{2})(\frac{1}{2} - \frac{1}{2} + 1)} \left| \frac{1}{2} - \frac{1}{2}, 11 \right\rangle + \sqrt{(1+1)(1-1+1)} \left| \frac{1}{2} \frac{1}{2}, 10 \right\rangle \right) \\ &= \frac{1}{\sqrt{3}} \left| \frac{1}{2} - \frac{1}{2}, 11 \right\rangle + \frac{\sqrt{2}}{\sqrt{3}} \left| \frac{1}{2} \frac{1}{2}, 10 \right\rangle. \end{aligned} \quad (2)$$

Then $\left| \frac{1}{2} \frac{1}{2} \right\rangle$ is the state orthogonal to $\left| \frac{3}{2} \frac{1}{2} \right\rangle$, so

$$\left| \frac{1}{2} \frac{1}{2} \right\rangle = \frac{1}{\sqrt{3}} \left| \frac{1}{2} \frac{1}{2}, 10 \right\rangle - \frac{\sqrt{2}}{\sqrt{3}} \left| \frac{1}{2} - \frac{1}{2}, 11 \right\rangle, \quad (3)$$

where the sign is chosen to follow rule (4) on p. 413 of the text. From eqns. (1), (2), and (3) we read off the Clebsch-Gordon coefficients

$$\begin{aligned} \langle \frac{1}{2} \frac{1}{2}, 11 | \frac{3}{2} \frac{3}{2} \rangle &= 1, \\ \langle \frac{1}{2} - \frac{1}{2}, 11 | \frac{3}{2} \frac{1}{2} \rangle &= \frac{1}{\sqrt{3}}, \quad \langle \frac{1}{2} - \frac{1}{2}, 11 | \frac{1}{2} \frac{1}{2} \rangle = -\sqrt{\frac{2}{3}}, \\ \langle \frac{1}{2} \frac{1}{2}, 10 | \frac{3}{2} \frac{1}{2} \rangle &= \sqrt{\frac{2}{3}}, \quad \langle \frac{1}{2} \frac{1}{2}, 10 | \frac{1}{2} \frac{1}{2} \rangle = \frac{1}{\sqrt{3}}. \end{aligned}$$

Then rule (5) on p. 413 gives the rest:

$$\begin{aligned} \langle \frac{1}{2} - \frac{1}{2}, 1-1 | \frac{3}{2} - \frac{3}{2} \rangle &= 1, \\ \langle \frac{1}{2} \frac{1}{2}, 1-1 | \frac{3}{2} - \frac{1}{2} \rangle &= \frac{1}{\sqrt{3}}, \quad \langle \frac{1}{2} \frac{1}{2}, 1-1 | \frac{1}{2} - \frac{1}{2} \rangle = \sqrt{\frac{2}{3}}, \\ \langle \frac{1}{2} - \frac{1}{2}, 10 | \frac{3}{2} - \frac{1}{2} \rangle &= \sqrt{\frac{2}{3}}, \quad \langle \frac{1}{2} - \frac{1}{2}, 10 | \frac{1}{2} - \frac{1}{2} \rangle = -\frac{1}{\sqrt{3}}. \end{aligned}$$

All others are zero.

(2): $1 \otimes 1 = 2 \oplus 1 \oplus 0$, so $j_1 = j_2 = 1$ implies $j \in \{2, 1, 0\}$. Since there is a unique state with $m = 2$ we have

$$|22\rangle = |11, 11\rangle. \quad (4)$$

Since $J_-|22\rangle = 2\hbar|21\rangle$,

$$|21\rangle = \frac{1}{2\hbar}J_-|22\rangle = \frac{1}{2\hbar}(J_{1-} + J_{2-})|11, 11\rangle = \frac{1}{\sqrt{2}}|10, 11\rangle + \frac{1}{\sqrt{2}}|11, 10\rangle. \quad (5)$$

Since $J_-|21\rangle = \sqrt{6}\hbar|20\rangle$,

$$\begin{aligned} |20\rangle &= \frac{1}{\sqrt{6}\hbar}(J_{1-} + J_{2-})(|10, 11\rangle + |11, 10\rangle)\frac{1}{\sqrt{2}} \\ &= \frac{1}{2\sqrt{3}\hbar}(J_{1-}|10, 11\rangle + J_{2-}|10, 11\rangle + J_{1-}|11, 10\rangle + J_{2-}|11, 10\rangle) \\ &= \frac{1}{2\sqrt{3}}(\sqrt{2}|1-1, 11\rangle + \sqrt{2}|10, 10\rangle + \sqrt{2}|10, 10\rangle + \sqrt{2}|11, 1-1\rangle) \\ &= \frac{1}{\sqrt{6}}|1-1, 11\rangle + \sqrt{\frac{2}{3}}|10, 10\rangle + \frac{1}{\sqrt{6}}|11, 1-1\rangle. \end{aligned} \quad (6)$$

$|11\rangle$ is the state orthogonal to $|21\rangle$, so

$$|11\rangle = \frac{1}{\sqrt{2}}|11, 10\rangle - \frac{1}{\sqrt{2}}|10, 11\rangle, \quad (7)$$

and $J_-|11\rangle = \sqrt{2}\hbar|10\rangle$ so

$$\begin{aligned} |10\rangle &= \frac{1}{\sqrt{2}\hbar}(J_{1-} + J_{2-})(|11, 10\rangle - |10, 11\rangle)\frac{1}{\sqrt{2}} \\ &= \frac{1}{2\hbar}(J_{1-}|11, 10\rangle + J_{2-}|11, 10\rangle - J_{1-}|10, 11\rangle - J_{2-}|10, 11\rangle) \\ &= \frac{1}{2}(\sqrt{2}|10, 10\rangle + \sqrt{2}|11, 1-1\rangle - \sqrt{2}|1-1, 11\rangle - \sqrt{2}|10, 10\rangle) \\ &= \frac{1}{\sqrt{2}}|11, 1-1\rangle - \frac{1}{\sqrt{2}}|1-1, 11\rangle. \end{aligned} \quad (8)$$

Finally, $|00\rangle$ is perpendicular to both $|20\rangle$ and $|10\rangle$. Say $|00\rangle = a|11, 1-1\rangle + b|10, 10\rangle + c|1-1, 11\rangle$. Then

$$\begin{aligned} 0 &= \langle 00|20\rangle = \frac{1}{\sqrt{6}}(a + 2b + c), \\ 0 &= \langle 00|10\rangle = \frac{1}{\sqrt{2}}(a - c), \end{aligned}$$

which together imply $a = -b = c$. Then normalizing $|00\rangle$ implies $a = 1/\sqrt{3}$, so

$$|00\rangle = \frac{1}{\sqrt{3}}|11, 1-1\rangle - \frac{1}{\sqrt{3}}|10, 10\rangle + \frac{1}{\sqrt{3}}|1-1, 11\rangle. \quad (9)$$

From eqns. (4–9) by taking inner products and by using rule (5) on p. 413 we get all the Clebsch-Gordon coefficients.

Problem 15.2.3:

$$\begin{aligned} \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} &= \frac{1}{2} \otimes \left(\frac{1}{2} \otimes \frac{1}{2}\right) = \frac{1}{2} \otimes (1 \oplus 0) = \left(\frac{1}{2} \otimes 1\right) \oplus \left(\frac{1}{2} \otimes 0\right) = \left(\frac{3}{2} \oplus \frac{1}{2}\right) \oplus \frac{1}{2} \\ &= \frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2}. \end{aligned}$$